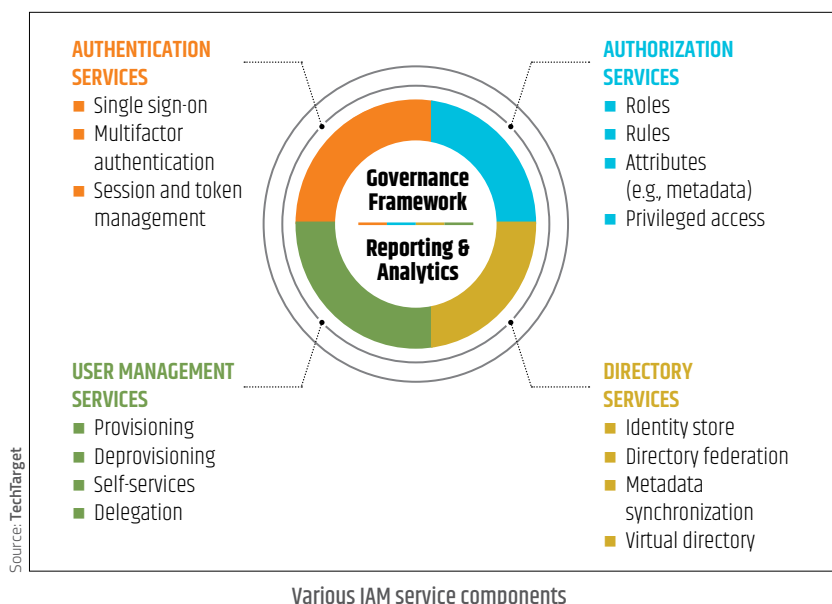




Identity and access management

(IAM): Identity and access management products track the user and what exactly they are allowed to access.

In accordance to this, IAM products either authorise users or deny access to unauthorised users as deemed fit. IAM is extremely important in cloud computing since the user's identity and access privileges dictate whether a user can access the data, not their device or location. IAM solutions are especially effective against account takeovers and insider attacks. IAM solutions usually include several services (separate or all-in-one) such as identity provider (IdP), single sign-on (SSO), multi-factor authentication (MFA), access control services and more.

Firewall: A cloud firewall provides protection around cloud data by blocking malicious traffic on the web. Cloud firewalls are hosted within the cloud and act as virtual barriers around the cloud infrastructure. Cloud firewalls prevent DDoS attacks, malicious bots and other vulnerability exploits.

WHAT DOES A CLOUD SECURITY ENGINEER DO?

A cloud security engineer or a cloud security specialist specialises in providing security for cloud-based platforms and plays an important role in safeguarding a cloud-based organisation's

data. A cloud security engineer's role often entails analysing existing cloud structures and attempting to come up with new, enhanced security solutions.

A typical cloud security engineer is expected to have a thorough understanding of the technologies we discussed in the previous section. In addition to those, they must also understand how to properly configure security settings for cloud servers to avoid exposed data. They must also perform threat simulations to detect possible risks and flaws in the cloud-based security programs of the organisation. Depending on the result of these simulations, cloud security engineers must be able to provide astute recommendations to improve security. It is their job to not only create security technologies for the cloud but also investigate, recommend and educate (users as well as employees). A cloud security engineer must also ensure consistency in the security policies and measures they implement across a company's infrastructure including public clouds, private clouds and on-premises infrastructure. If there are any inconsistencies, hackers will find it easier to target it.

QUALIFICATIONS AND SKILLS REQUIRED

Prospective candidates aspiring to become cloud security engineers must at

least have a bachelor's degree in fields such as computer science, programming, or information security. However, some firms do require a master's degree. Aspirants can also pursue certification through the Cloud Security Alliance (recently opened up its second Chapter in India) or the (ISC)².

Cloud security engineers need to possess strong technical skills such as experience with Windows OS or Linux, languages such as Python, micro-service development using Java and Node.js, knowledge of DevOps and DevOps tools and experience with cloud providing ecosystems such as AWS. Additionally, extraordinary attention to detail can help them succeed since they have to keep monitoring cloud systems to pick up on risks and threats.

SCOPE FOR THE FUTURE

According to a 2018 report by edtech platform, Great Learning, India is expected to see a massive rise in cloud computing jobs by 2022, thereby significantly increasing the demand for cloud security engineers. As more and more organisations are beginning to shift their operations to the cloud, the Indian cloud computing market is expected to grow to \$4 billion by the end of 2020. The field has also recorded an annual growth rate of over 30 per cent.

The prospects for cloud security engineers are bright since cloud engineers get paid a much higher salary package than traditional IT services roles in India, with entry-level salaries varying between ₹5 lakh to ₹7 lakh per annum. Also, cloud engineers with about 5 years of experience typically begin earning between ₹12 lakh to ₹19 lakh per annum. The cloud computing industry, not only in India but the entire world, witnesses a severe dearth of talent with over 1.7 million cloud jobs remaining vacant worldwide. According to IDC data, there is only one qualified candidate for every 100 job postings in cloud computing. Therefore, the need for skilled and talented cloud engineers is growing rapidly in India since the technology is also spreading its reach further in the country.